

Test I

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) You may not use a calculator for this test.
- V) Each problem is worth four points for a total of 100 points.

① Multiply.

$$6x^3(3x^4 - 2x^2 + 10x - 1)$$

② Determine if the expressions below are polynomials. If not, circle the term(s) that disqualify each expression.

i) $\tan x^2 - 4x + 2$

ii) $x^3 + \pi x - 9$

iii) $-9x^2 - \frac{2}{x^2} - 4x + 8.1$

iv) $\log(4x) + x^3 - x^2 - x - 1$

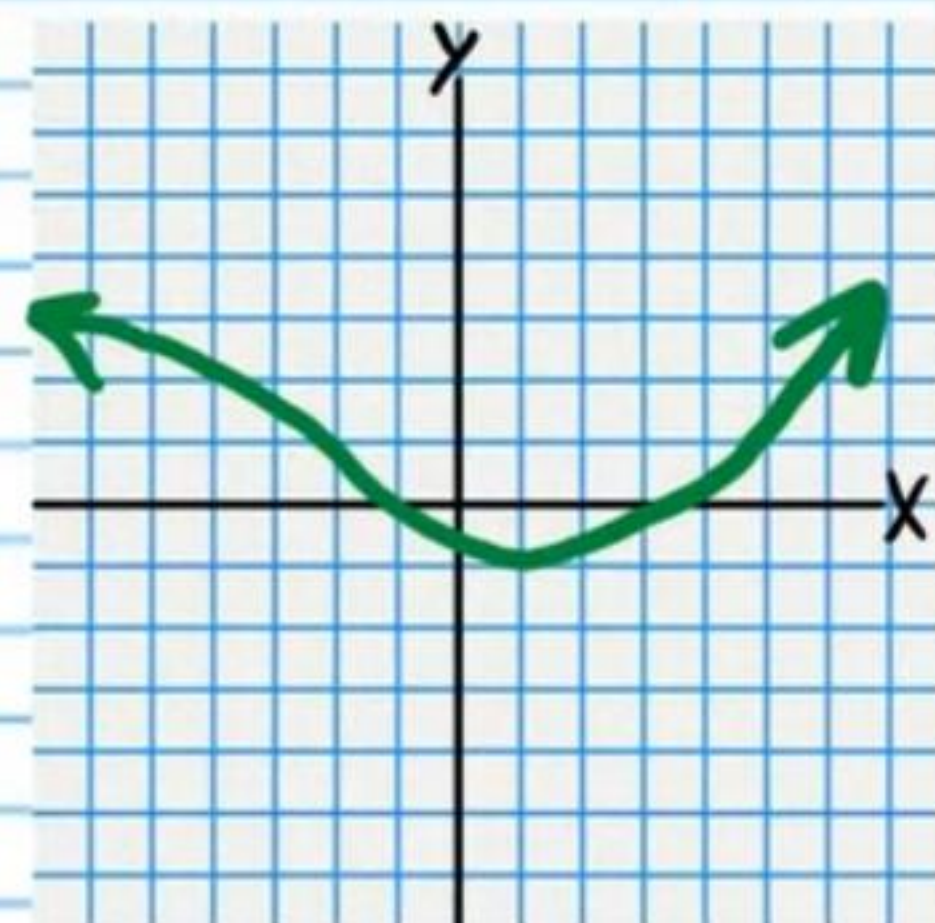
③ Indicate the end behavior of the function below.

$$f(x) = -7x^7 - 10x^6 + 3x^5 - 2x^4 - x^3 - x^2 + 8x + 17$$

④ The graph of a function is written below. Write the intervals where the function is increasing or decreasing.

Increasing:

Decreasing:



⑤ Use the Rational Zeros Theorem to find all of the potential rational solutions of the polynomial function.

$$f(x) = -6x^3 + x^2 - 8x + 7$$

⑥ What is the maximum number of turning points of the graph of the function below?

$$f(x) = 6x^4 - 2x^3 + x^2 - x + 1$$

⑦ Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

$$-4i, 3, 2 \quad LC=3 \quad D=4$$

⑧ A polynomial function is written below in factored form. Write the x-intercepts of the corresponding graph and determine if the function crosses or touches the x-axis at each intercept.

$$f(x) = \frac{1}{2}(x-5)^4(x+4)^4(x-3)^2$$

⑨ Divide using synthetic division.

$$(-2x^3 + x^2 + 3x - 8) \div (3 + x)$$

⑩ Solve. Write the solutions graphically and in interval notation.

$$-1 \leq 27x^3$$

$$\textcircled{11} \frac{x-1}{x+1} > 3$$

$$\textcircled{12} 5x + 14 > \frac{3}{x}$$

$$\textcircled{13} x^3 + 10 \leq 4x^2 + 7x$$

⑭ Find the x-intercepts of the graph of the function below and determine if the graph crosses or touches the x-axis at each intercept. ~~★~~ Hint: Imaginary zeros are not x-intercepts.

$$f(x) = x^3 - 2x^2 + 4x - 8$$

⑮ Find the zeros of the function below. Write all of them, whether rational, irrational, or imaginary.

$$f(x) = 3x^4 - 22x^3 - 73x^2 - 68x - 20$$

⑩ Find the roots of the polynomial function below, whether rational, irrational, or imaginary.

$$f(x) = x^4 - 4x^3 - 20x^2 - 4x - 21$$

⑪ Find the solutions of the equation below, whether rational or irrational.

$$0 = 4x^5 - 4x^4 - 13x^3 + 20x^2 - 5x - 2$$

⑮ Divide.

$$(x^5 - x^3 - 2x) \div (3x - 2 + x^2)$$

⑯ Divide using synthetic division.

$$(x^5 + 3x^4 + 10x^2 + 1) \div (x + 4)$$

⑰ Find the zeros of the polynomial function below.
Write the function in factored form.

$$f(x) = x^3 - 5x^2 + 3x + 9$$

② Find the roots of the polynomial below, whether rational, irrational, or imaginary.

$$f(x) = x^4 - 2x^2 - 80$$

Graph the following functions and find all of the information requested. Draw asymptotes as dotted lines.

$$\textcircled{2d} f(x) = \frac{-4x^2 - 5x + 9}{x^2 + 2x - 3}$$

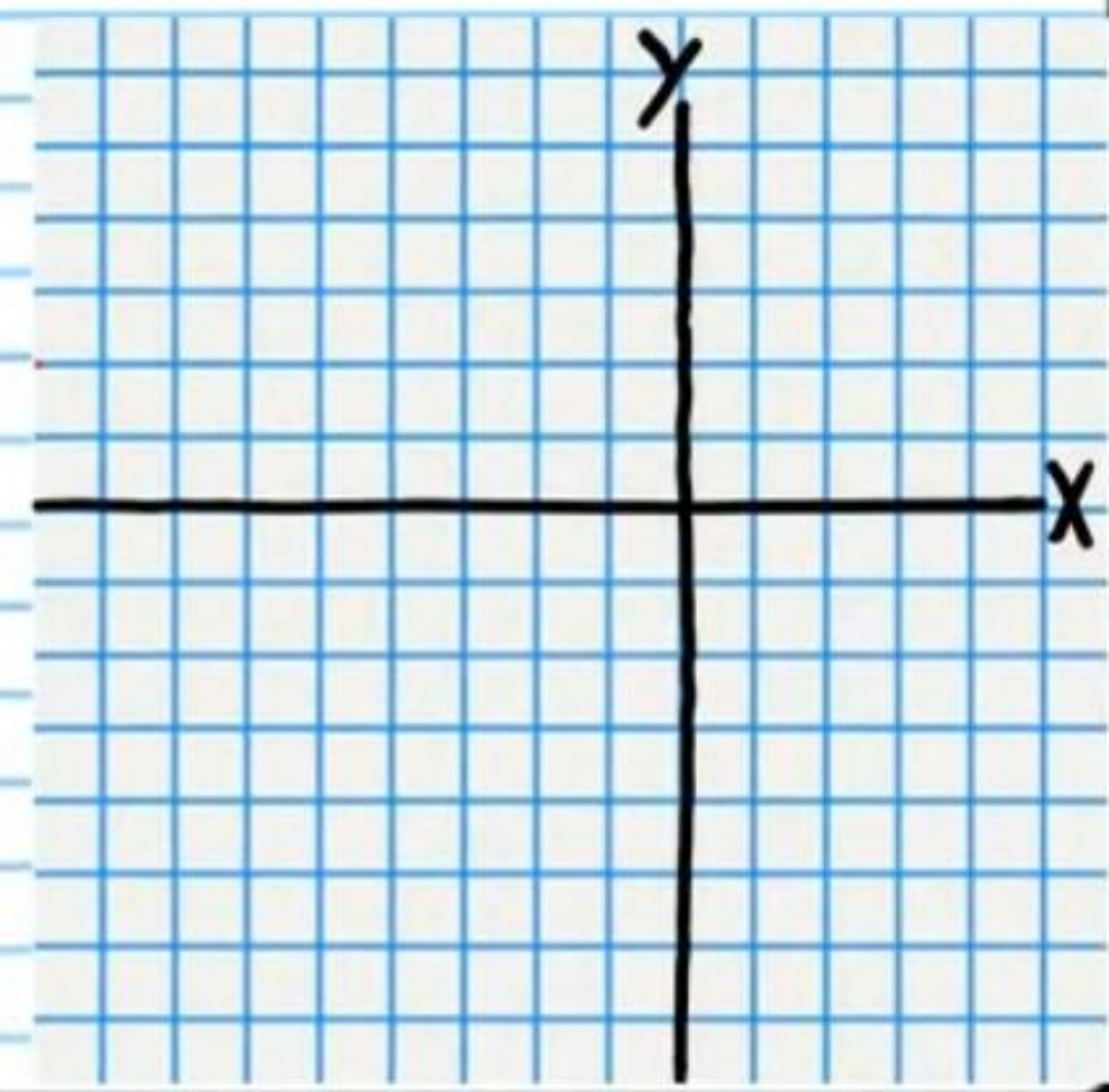
Horizontal Asymptote Equation

Hole(s)

Vertical Asymptote Equation(s) Sign Chart

Y-Intercept

X-Intercept



②③ $f(x) = \frac{x+4}{x^2+5x+4}$

Hole(s)

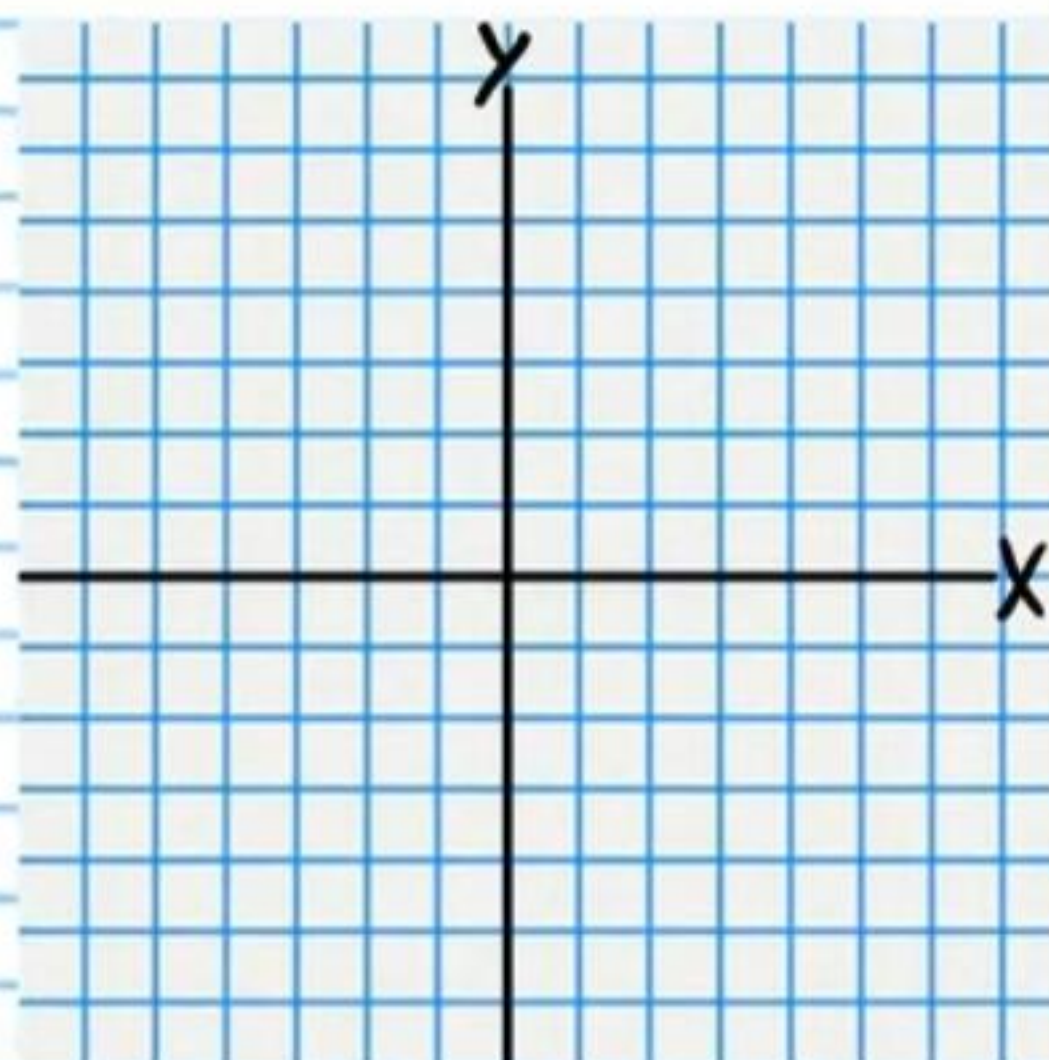
Vertical Asymptote Equation(s)

Horizontal Asymptote Equation(s)

Sign Chart

Y-Intercept

X-Intercept



②④ $f(x) = \frac{x-2}{x^2-x-6}$

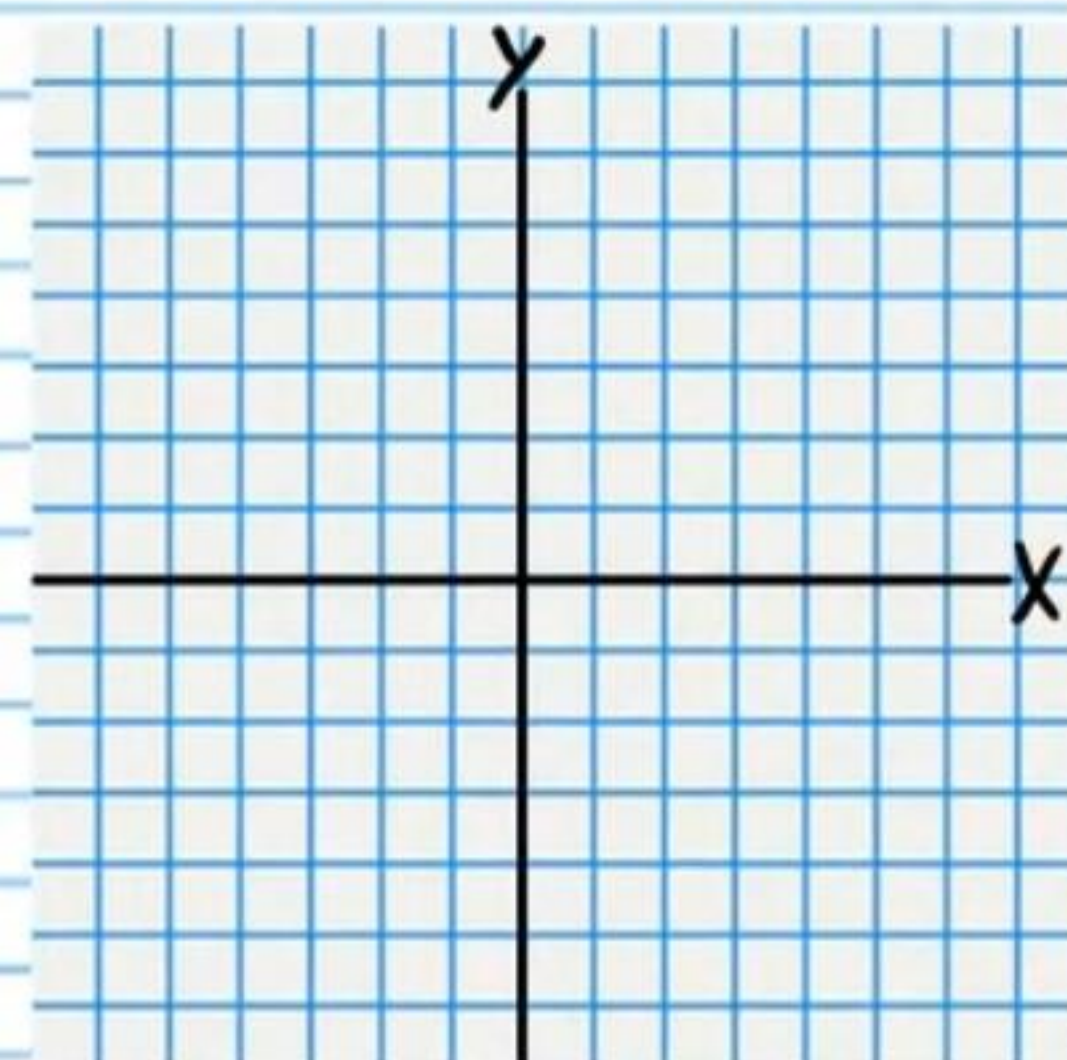
Vertical Asymptote Equation(s)

Horizontal Asymptote Equation(s)

Sign Chart

Y-Intercept

X-Intercept



* Hint:
Factor a
GCF.

(25) $f(x) = \frac{x^4 - 16}{x^3 + x^2 - 4x - 4}$ Oblique Asymptote Equation

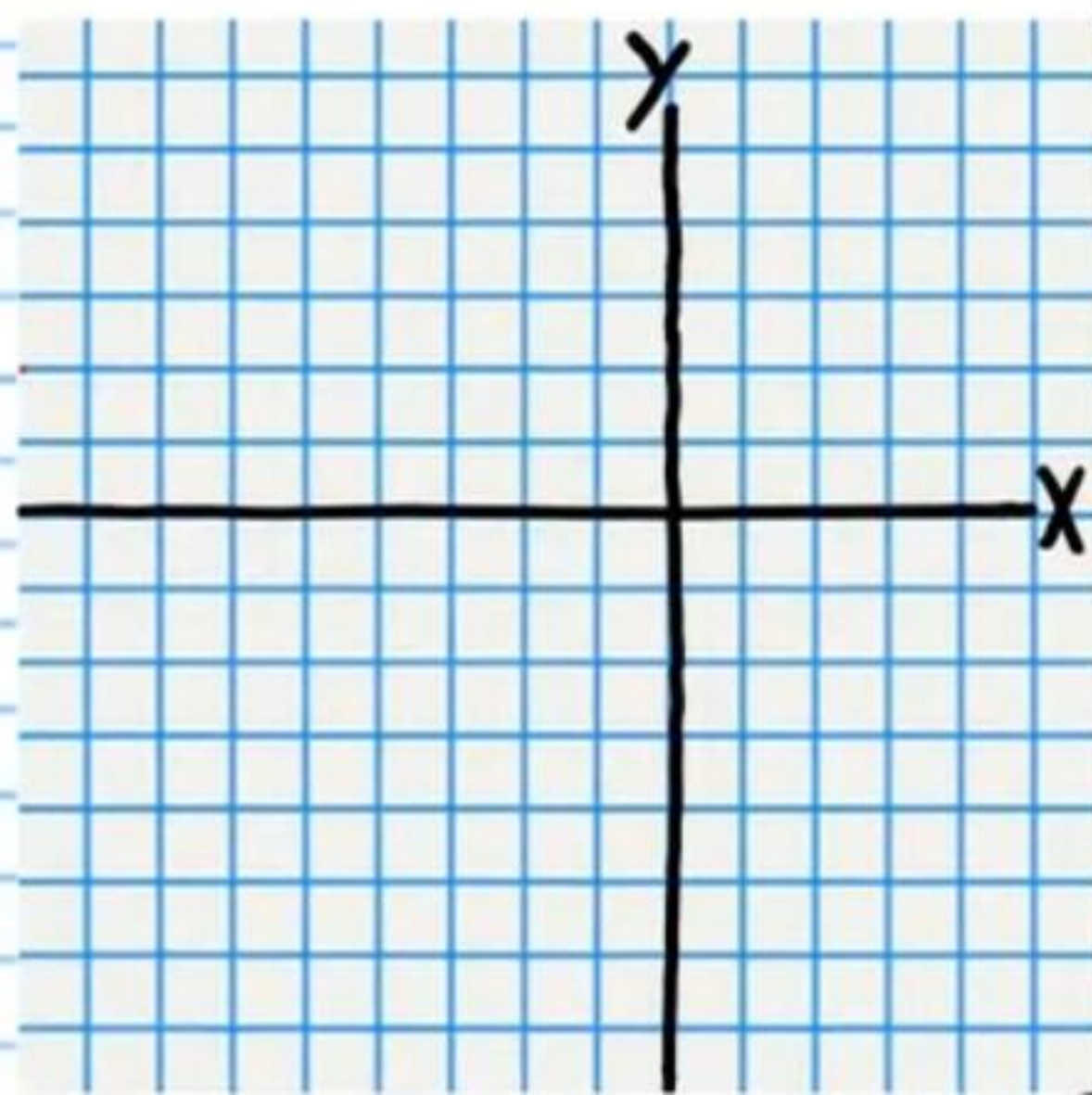
Hole(s)

Vertical Asymptote Equation(s)

Sign Chart

Y-Intercept

X-Intercept



Answers

① $18x^7 - 12x^5 + 60x^4 - 6x^3$

② i) $\tan x^2 - 4x + 2$

No

ii) $x^3 + \pi x - 9$

Yes

iii) $-9x^2 - \frac{2}{x^2} - 4x + 8.1$

No

iv) $\log(4x) + x^3 - x^2 - x - 1$

No

③ As $x \rightarrow \infty, f(x) \rightarrow -\infty$
As $x \rightarrow -\infty, f(x) \rightarrow \infty$

④ Increasing: $(1, 7)$

Decreasing: $(-7, 1)$

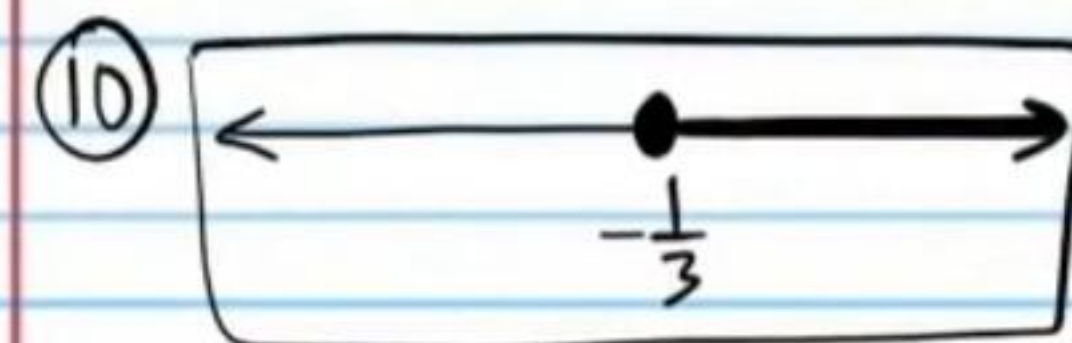
⑤ $\pm 1, \pm 7, \pm \frac{1}{2}, \pm \frac{7}{2}, \pm \frac{1}{3}, \pm \frac{7}{3}, \pm \frac{1}{6}, \pm \frac{7}{6}$

⑥ 3

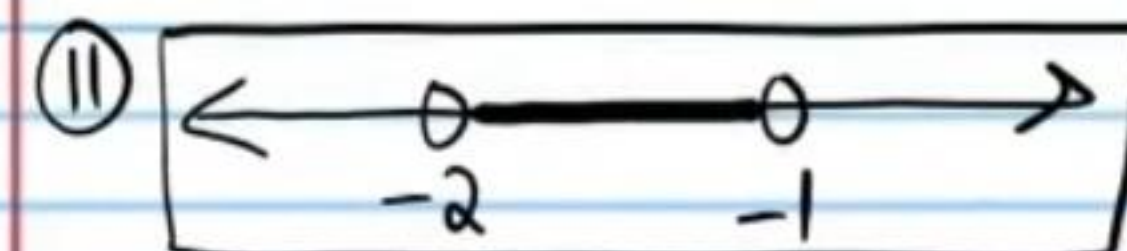
⑦ $3x^4 - 15x^3 + 66x^2 - 240x + 288$

⑧ $5 \Rightarrow$ Touches
 $-4 \Rightarrow$ Touches
 $3 \Rightarrow$ Touches

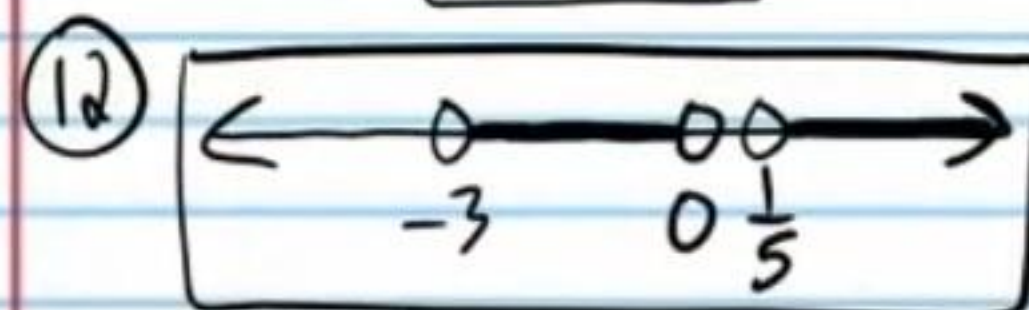
⑨ $-2x^2 + 7x - 18 + \frac{46}{x+3}$



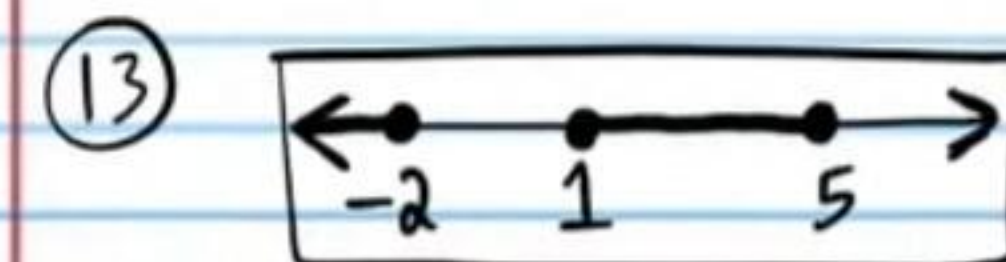
$[-\frac{1}{3}, \infty)$



$(-2, -1)$



$(-3, 0) \cup (\frac{1}{5}, \infty)$



$(-\infty, -2] \cup [1, 5]$

⑭ $2 \Rightarrow$ Crosses

⑮ $-\frac{2}{3}, -1, 10$

①6 $i, -i, 7, -3$

①7 $1, -2, \frac{1+\sqrt{2}}{2}, \frac{1-\sqrt{2}}{2}$

①8 $x^3 - 3x^2 + 10x - 36 + \frac{126x - 72}{x^2 + 3x - 2}$

①9 $x^4 - x^3 + 4x^2 - 6x + 24 - \frac{95}{x+4}$

②0 $-1, 3$ $f(x) = (x+1)(x-3)^2$ or $f(x) = (x+1)(x-3)(x-3)$

②1 $\sqrt{10}, -\sqrt{10}, 2i\sqrt{2}, -2i\sqrt{2}$

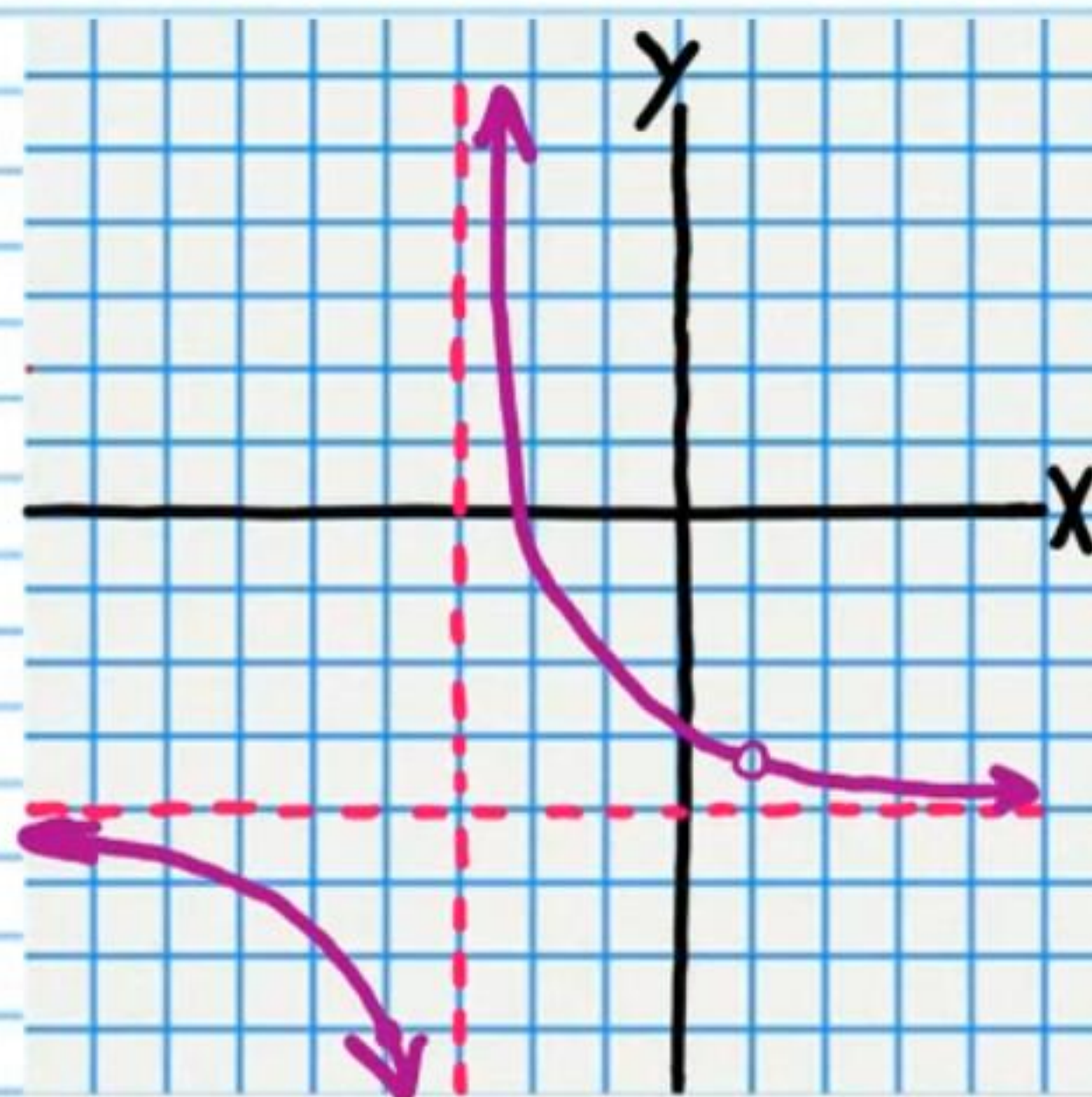
②2 HA: $y = -4$

VA: $x = -3$

Hole: $x = 1$

X-Intercept: $(-\frac{9}{4}, 0)$

Y-Intercept: $(0, -3)$



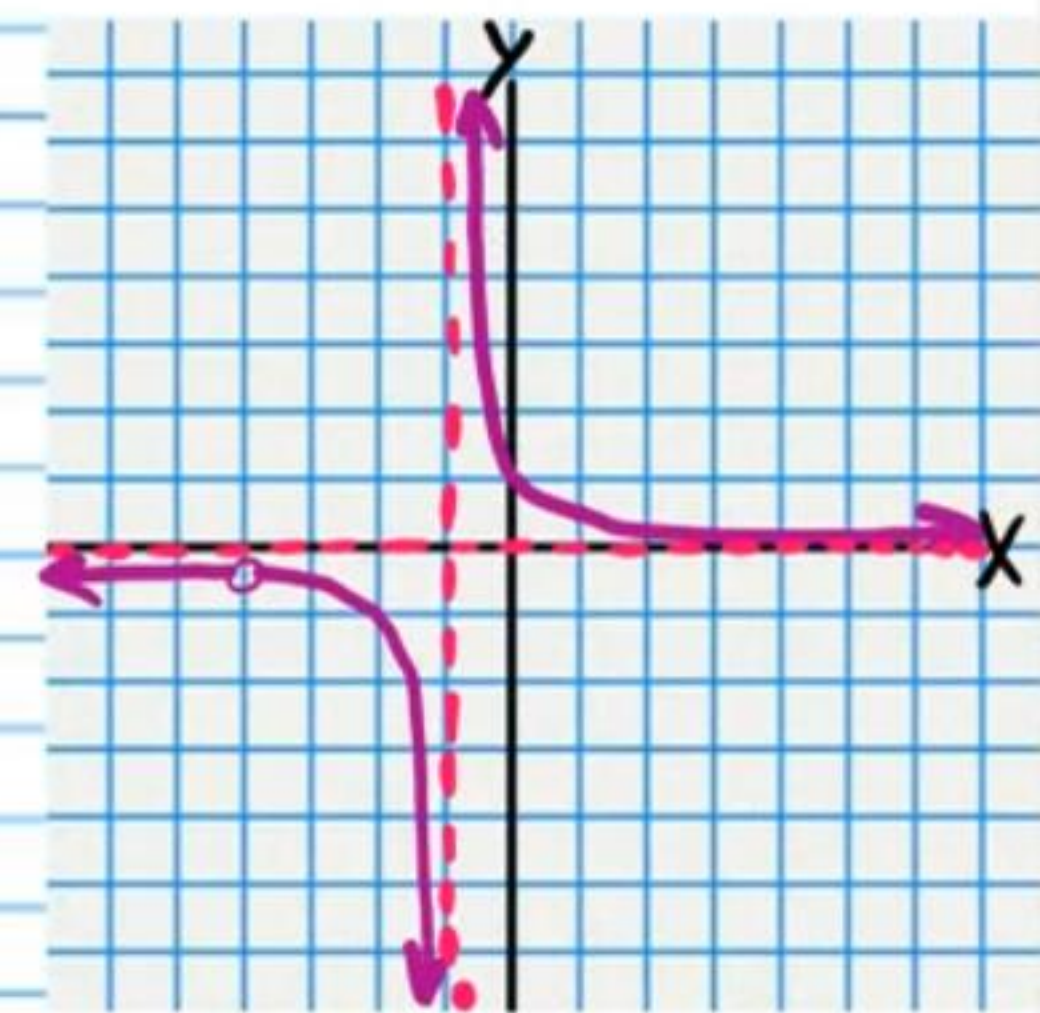
②3 Hole: $x = -4$

VA: $x = -1$

HA: $y = 0$

Y-Intercept: $(0, 1)$

X-Intercept: None

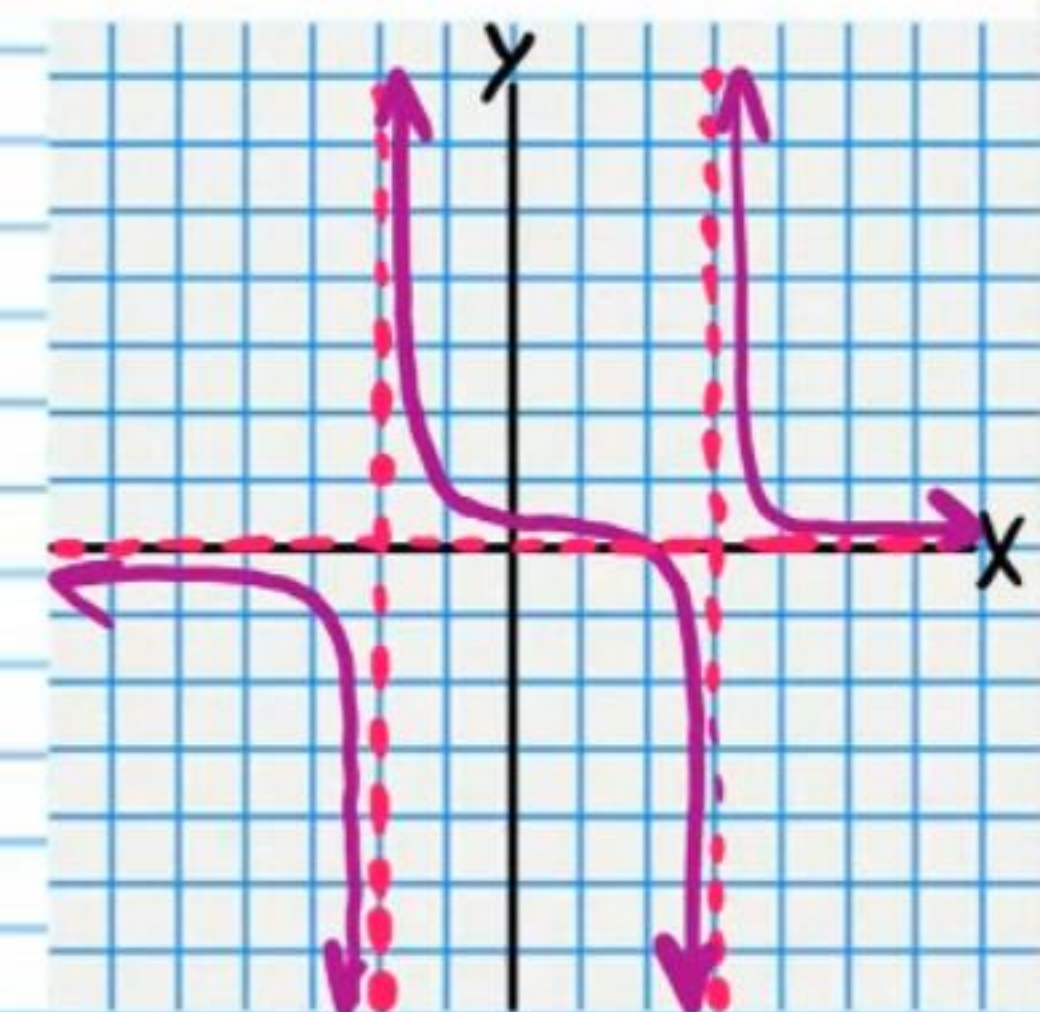


②4 VA: $x = 3, x = -2$

HA: $y = 0$

X-Intercept: $(2, 0)$

Y-Intercept: $(0, \frac{1}{3})$



②5 OA: $y = x - 1$

Hole: $x = 2, x = -2$

VA: $x = -1$

Y-Intercept: $(0, 4)$

X-Intercept: None

